

worked_example

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$y \sim X + 1 + (x|subject)$
 $y = X\beta + Zu + \epsilon$
 $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$ (7x2)
 $Z = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$ (7x6)
 $\beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$ (2x1)
 $u = \begin{bmatrix} u_1 \\ u_2 \\ u_1 \\ u_2 \\ u_1 \\ u_2 \end{bmatrix}$ (6x1)
 $\epsilon \sim N(0, \sigma^2)$
 $n_{subject} = 3$
 $q = p \times n_{subject}$
 $u \sim N(0, G)$
 $G_{q \times q}$

$x_{n \times 1}$ is the independent variable from three different subjects, colour coded.

$y_{n \times 1}$ is the dependent variable, measured from the same subjects, colour coded.

$X_{n \times p}$ is the design matrix for the general effects across participants (first level effects) (intercept and slope)

$Z_{q \times p}$ is the design matrix for the effects for each participant (second level effects) (intercept and slope for each subject)

$\beta_{p \times 1}$ are the coefficients for the two parameters of X

$u_{q \times 1}$ are the coefficients for the six parameters of Z ($q = n_{subjects} \times p$)

$\epsilon_{n \times 1}$ are the residuals, assumed to normally distributed with a mean of 0 and a variance of σ^2

u is normally distributed with a mean of 0 and a variance-covariance structure $G_{q \times q}$